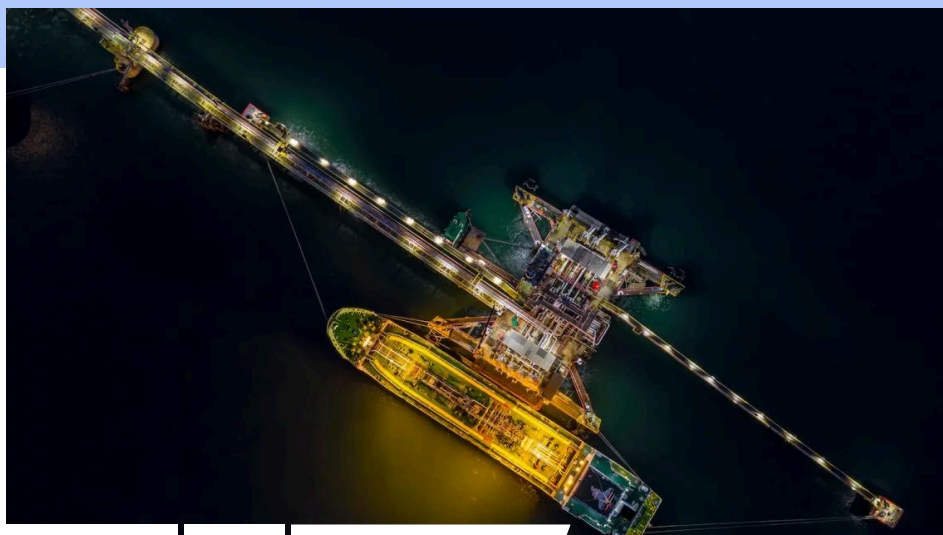




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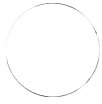
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Charting the uncharted: The (un)intended consequences of oil sanctions and dark shipping

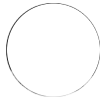
Jesús Fernández-Villaverde, Yiliang Li, Le Xu, Francesco Zanetti / 11 Mar 2025

Despite formal restrictions, untracked oil continues to reach global markets, calling into question the true effectiveness of oil sanctions. This column presents a novel ship clustering model to detect 'dark shipping' - where tankers disable their location signals to evade detection - linked to sanctioned oil transport. Following Western sanctions on Russia in 2022, officially recorded Russian oil exports declined, but shipments via dark vessels more than doubled. A significant share ultimately reaches non-sanctioning countries. The increased availability of discounted, dark-shipped oil helps keep global prices down.

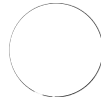
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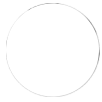
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In recent years, sanctions have become a frequently used tool for Western authorities to curb oil exports from targeted economies, aiming to reduce the latter's export revenues by restricting access to major markets. However, the broader consequences of such measures remain uncertain – particularly given mounting evidence that some tankers 'go dark' by disabling their location signals to evade detection and break the sanctions. Additionally, the effects on the global supply chain are often overlooked, raising further questions about the true impact of these restrictions.

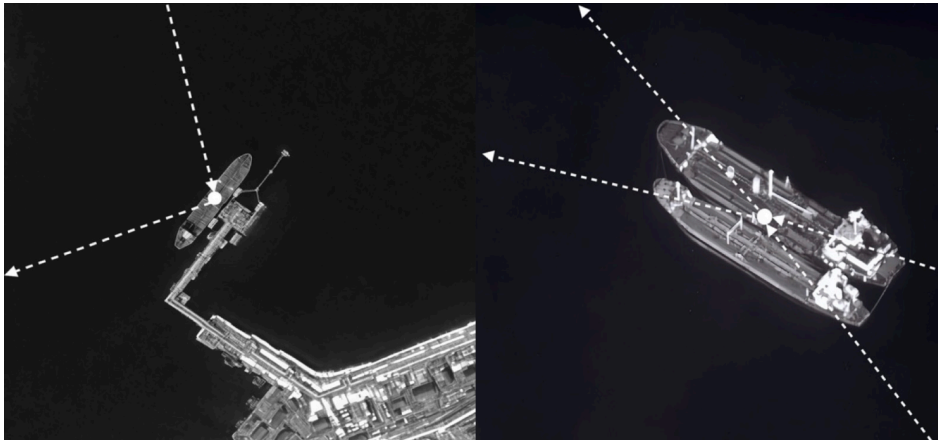
The unauthorised practice known as 'dark shipping' has gained increasing attention in the context of sanctions on countries such as Iran, Syria, Venezuela, and Russia (Babina et al. 2023, Laudati and Pesaran 2023, Kilian et al. 2024b, Rodríguez 2025). Despite formal restrictions, untracked oil continues to reach global markets, calling into question the true effectiveness of sanctions in curbing exports. These hidden shipments can, in turn, influence global oil supply, impact energy prices, and reshape trade patterns more significantly than official data alone might indicate.

In a recent paper (Fernández-Villaverde et al. 2025), we present a novel ship clustering model to detect dark shipping linked to sanctioned oil transport. By analysing vessel-specific characteristics, evasive behaviour during signal gaps, and navigational anomalies, we quantify unauthorised tanker shipments across the global crude oil fleet. Using this data, along with state-of-the-art methods for causal inference in time series, we investigate how sanctions and dark shipping affect oil flows, prices, and economic activity in both sanctioning and non-sanctioning countries.

What is dark shipping?

Dark shipping occurs when vessels intentionally disable or manipulate their automatic identification system (AIS), concealing their location, course, and speed from global tracking systems. These AIS gaps enable tankers to load or unload oil without official reporting, often in regions with weak sanction enforcement, such as international waters.

Figure 1 Satellite imagery of a direct visit to a suspicious port and a ship-to-ship transfer



Notes: The left panel displays satellite imagery of the dark ship Roma (IMO: 9182291) at Kharg Island, Iran, on 20 August 2022, during a period when its automatic identification system (AIS) transceiver was off. Dashed white arrows indicate the vessel's estimated trajectory during the data gap. The right panel shows satellite imagery of an unauthorised ship-to-ship transfer between two dark ships, Abyss (IMO: 9157765) and Shanaye Queen (IMO: 9242118), on 28 January 2022 in the Persian Gulf, with dashed white arrows marking their respective trajectories. All three dark ships were detected using our ship clustering model.

Tankers may either load oil directly from sanctioned ports while their AIS signals are off or conduct ship-to-ship transfers between dark vessels (see Figure 1). By operating covertly, they capitalise on premiums offered by sanctioned producers striving to maintain exports. This mechanism explains how oil from restricted countries continues to reach global markets despite sanctions.

With seaborne transport representing over three-quarters of global crude oil trade (US Energy Information Administration 2024), these evasive tactics can significantly distort official trade statistics. Our research offers a systematic approach to identifying these covert activities and assessing their broader economic impact.

A ship clustering model to identify dark ships

Our study employs a multi-attribute ship clustering model based on extensive AIS records covering the global crude oil tanker fleet from 2017 to 2023. Instead of relying on a single attribute, such as vessel age, the model integrates three key layers of analysis:

1. **Vessel characteristics**, including the tanker's age, the number of vessels operated by the same company, and flag registration. We find that vessels with 'flags of convenience' (i.e. flags from jurisdictions with lower oversight) and older ships are more likely to exhibit suspicious behaviours.
2. **Trip-level information**. We track each tanker's voyages from port to port, noting origins, destinations, and whether it appears near sanctioned countries during an AIS gap. This trip-level view clarifies when a tanker is likely to be picking up or delivering cargo without proper disclosure.
3. **Movement dynamics**. The model also analyses the frequency, duration, and timing of AIS dark periods, along with anomalies in speed or heading. Crucially, it identifies potential ship-to-ship transfers by detecting instances where two vessels go dark simultaneously while in close geographical proximity.

By integrating these features, we assign a 'dark score' to each vessel and trip. A tanker that rarely goes dark and avoids sanctioned regions during AIS gaps is classified as 'white', indicating a low likelihood of involvement in covert oil transport. Conversely, a vessel with multiple suspicious voyages, prolonged AIS gaps, and routes closely aligned with sanctioned producers is flagged as 'dark'.

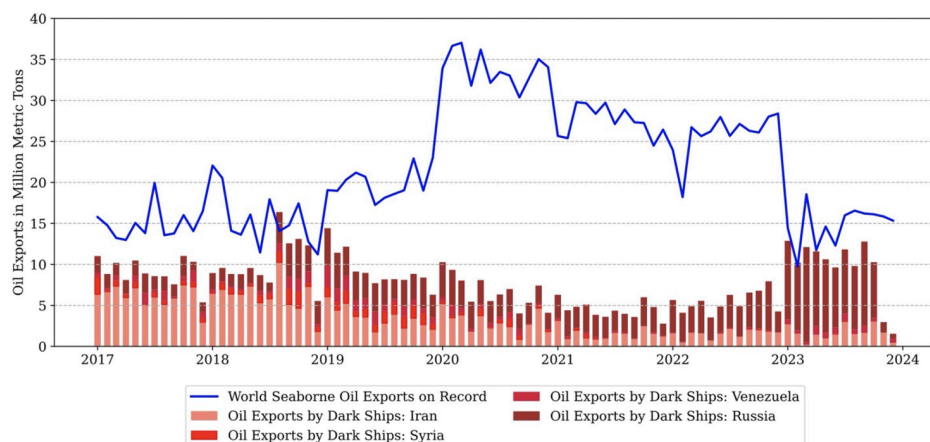
This approach minimises dependence on external data, such as satellite imagery or ownership records, which may be incomplete or costly. Instead, it leverages widely available AIS data and a structured methodology to systematically detect dark shipping across the global fleet.

Tracking the flows of dark-shipped oil

We find that dark shipping is far from a marginal activity. As shown in Figure 2, from 2017 to 2023, an average of 7.8 million metric tonnes of crude oil per month were exported via dark ships from Iran, Russia, Syria, and Venezuela – equivalent to approximately 43% of total global seaborne crude exports reported in official UN Comtrade statistics.

The prevalence of dark shipping also rises significantly following major policy shifts, such as the US withdrawal from the Joint Comprehensive Plan of Action in 2018 and the imposition of oil embargoes and price caps on Russian exports in 2022–2023.

Figure 2 Oil exports of sanctioned countries by dark ships vs world seaborne oil exports on record



Notes: The stacked bars represent estimated monthly crude oil exports transported by dark ships from Iran, Russia, Syria, and Venezuela. The solid blue line shows global seaborne crude oil exports, sourced from the UN Comtrade database (HS code 2709: petroleum oils and oils from bituminous minerals, crude). Both series are measured in million metric tonnes.

From late 2022 onward, data on Russia suggests that while officially recorded exports declined following Western sanctions, shipments via dark vessels more than doubled. This rechannelling of oil flows appears to have offset much of the expected supply contraction. By effectively increasing oil supply, these untracked shipments may exert downward pressure on global oil prices.

Another key insight is that a significant share of this oil ultimately reaches non-sanctioning countries (e.g. India), often at discounted prices (Hilgenstock et al. 2023, Kilian et al. 2024a). This realignment underscores how oil can bypass formal restrictions when substantial buyers remain outside the sanctioning coalition.

(Un)intended macroeconomic effects

To assess the impact of these flows on global economic activity, we use local projections (Jordà 2005, Barnichon and Brownlees 2019) to analyse how changes in sanctioned oil flows correlate with shifts in industrial production and prices in the US, the EU, and China.

60%

For the US, as a net oil exporter, the increased availability of discounted, dark-shipped oil helps keep global prices down. In the short run, this depresses US oil production and related sectors, as lower global prices reduce revenues. However, US consumers and downstream industries benefit from cheaper imported goods – particularly from China, where lower energy costs enhance manufacturing competitiveness. The net effect is mixed: a short-term contraction in US energy-related industries, offset by long-term gains in broader non-energy sectors due to lower input costs from non-sanctioning countries.

As a net oil importer, the EU faces higher energy costs if sanctions temporarily drive up recorded global oil prices. However, rising export demand from China – whose production benefits from cheaper energy prices – provides a boost to EU manufacturers. The result is a complex dynamic of demand-driven inflation and output expansion, where some sectors benefit from China's growth while others struggle with higher energy costs.

Access to discounted oil imports lowers energy costs in China, stimulating domestic production. This, in turn, boosts industrial output in both the US and the EU through increased trade, as Chinese manufacturers benefit from cheaper inputs and heightened competitiveness.

Relevance for the future

Our research highlights how oil sanctions have far-reaching aggregate effects beyond the targeted economies, often unexpectedly benefiting countries engaged in global trade. These insights are crucial for policymakers, who must account for the intended and unintended consequences of dark shipping on the global supply chain when designing and adapting oil-related sanctions.

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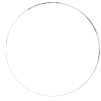


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